Ultimate Notes: Python for Data Analytics

1. What is Python

Python is a high-level, interpreted programming language known for its simplicity and readability.

Features of Python

1. Easy syntax
2. Open-source
3. Cross-platform
4. Extensive libraries

Data Types in Python

1. int, float, str, bool, list, tuple, set, dict

Advantages

1. Easy to learn
2. Large community
3. Versatile

Disadvantages

1. Slower execution
2. More memory usage

Python in Data Analytics

Python is used to analyze, visualize, and process data.

Advantages

1. Handles large data
2. Powerful libraries
3. Automation

Limitations

1. Performance issues in very large systems

Installation (Python, Jupyter, VS Code, PyCharm)

1. Download Python from python.org
2. Install and add to PATH
3. Install Anaconda for Jupyter Notebook
4. Install VS Code and add Python extension
5. Install PyCharm IDE

NumPy Library

What is NumPy

NumPy is used for numerical operations and array processing.

Features

1. Fast calculations
2. Supports arrays
3. Mathematical functions

Arithmetic Functions

```
np.add(arr,2)
np.subtract(arr,2)
np.multiply(arr,2)
```

Statistical Functions

```
np.mean(arr)
np.std(arr)
np.var(arr)
```

Mathematical Functions

```
np.sqrt(arr)
np.log(arr)
np.exp(arr)
```

Pandas Library

What is Pandas

Used for data manipulation and analysis.

Features

1. DataFrames
2. Handles missing data
3. Flexible

Functions

```
df.head()
df.info()
df.describe()
df.dropna()
df.groupby('col').mean()
```

Matplotlib Library

What is Matplotlib

Used for plotting graphs.

Functions

```
plt.plot(x,y)
plt.bar(x,y)
plt.hist(data)
plt.show()
```

Seaborn Library

What is Seaborn

Advanced visualization library.

Functions

```
sns.histplot(data)
sns.countplot(x=data)
sns.heatmap(data)
```